Matematica - Clasa a 6-a

Exercitii: Capitol I - Multimi si numere naturale

Nume: ________________________

Total: 100p + 10p oficiu

Data: ________________________

Ex. 1. Operatii cu multimi: (12p)

Fie $A = \{1,3,5,7,9\}$, $B = \{1,2,3,4,5\}$, $C = \{2,4,6,8,10\}$.

Calculeaza:

a) $A \cup B =$ ____________________________ b) $A \cap B =$ ____________________________

c) $A \setminus B =$ ____________________________ d) $B \setminus A =$ ____________________________

e) $A \cap C =$ ____________________________ f) $(A \cup B) \cap C =$ ____________________________

g) $A \cup (B \cap C) =$ ____________________________ h) $(A \setminus B) \cup (B \setminus A) =$ ________________________

Ex. 2. Submultimi: (8p)

a) Scrie toate submultimile lui $M = \{a, b, c\}$. (Sunt 2^3 = 8)

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b) Completeaza cu $\in$ sau $\notin$ sau $\subseteq$ sau $\not\subseteq$:

- 3 ___ $\{1,2,3,4\}$
- 3 ___ $\{1,2,3,4\}$
- $(1, 2)$ ___ $\{1,2,3\}$
- $\emptyset$ ___ $\{1,2,3\}$
- $(1, 5)$ ___ $\{1,2,3\}$
- 4 ___ $\{1,3,5\}$

Ex. 3. Criterii de divizibilitate: (10p)

a) Incercuieste cu ce numere se divide fiecare:

7.452 : 2 / 3 / 4 / 5 / 6 / 9 / 10

13.500 : 2 / 3 / 4 / 5 / 6 / 9 / 10

28.416 : 2 / 3 / 4 / 5 / 6 / 9 / 10

90.630 : 2 / 3 / 4 / 5 / 6 / 9 / 10

b) Gaseste cifra x astfel incat numarul sa fie divizibil cu 9:

5x31 : x = ___ 3x7x : x = ___ x45x2 : x = ___

Ex. 4. Numere prime si descompunere in factori primi: (15p)

a) Incercuieste numerele PRIME din lista:

1, 4, 7, 11, 15, 17, 21, 23, 27, 29, 33, 37, 41, 49, 51, 53

b) Descompune in factori primi (foloseste tabloul):

180:　　504:　　1260:　　2520:

180 = _________________________

504 = ___________________ 1260 = ________________________

2520 = ________________________

c) Calculeaza numarul de divizori:

nr. div(180) = ___ ; nr. div(504) = ___ ; nr. div(1260) = ___

Ex. 5. C.m.m.d.c. - toate metodele: (15p)

a) Prin listarea divizorilor comuni:

c.m.m.d.c.(24, 36):

D(24) = {__________________________________}

D(36) = {__________________________________}

Div. comuni = {____________} => c.m.m.d.c.(24, 36) = ___

b) Prin factori primi:

c.m.m.d.c.(360, 252, 180):

360 = _______ 252 = _______ 180 = _______

Factori comuni: _______ Puteri minime: _______

c.m.m.d.c.(360, 252, 180) = _______

c) Algoritmul lui Euclid:

c.m.m.d.c.(126, 84):

Exercitii: Capitol I (continuare)

Ex. 6. C.m.m.m.c. - calcule: (15p)

a) Prin listarea multiplilor:

c.m.m.m.c.(8, 12):

M(8) = 0, ___, ___, ___, ___, ___, ___, ___

M(12) = 0, ___, ___, ___, ___, ___, ___, ___

c.m.m.m.c.(8, 12) = ___

b) Prin factori primi:

c.m.m.m.c.(360, 252, 180):

360 = _______ 252 = _______ 180 = _______

Toti factorii: _______ Puteri maxime: _______

c.m.m.m.c.(360, 252, 180) = _______

c) Folosind formula a x b = cmmdc x cmmmc:

c.m.m.d.c.(48, 72) = 24. Calculeaza c.m.m.m.c.(48, 72).

48 × 72 = ___ ⇒ c.m.m.m.c. = ___ ÷ 24 = ___

d) Stim ca c.m.m.d.c.(a, b) = 15 si c.m.m.m.c.(a, b) = 180. Stiind ca a = 45, gaseste b.

45 × b = ___ × ___ ⇒ b = ___

Ex. 7. Proprietatile divizibilitatii: (10p)

a) Completeaza cu ADEVARAT sau FALS si justifica:

- Daca 6 / a si 6 / b, atunci 6 / (a + b) A / F => ___________
- Daca 4 / 24 si 4 / 10, atunci 4 / (24 - 10) A / F => ___________
- Daca 3 / a, atunci 3 / (5a) A / F => ___________
- Daca 5 / (a + b) atunci 5 / a si 5 / b A / F => ___________
- Daca 7 / (a × b) si 7 nu divide a, atunci 7 / b A / F => ___________

b) Arata ca 6 / (n{SUP2} - n) pentru orice n natural. (Indicatie: n{SUP2}-n = n(n-1)(n+1)/...)

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Ex. 8. Probleme aplicative: (15p)

a) 336 de mere si 252 de pere se pun in cosuri identice, fara rest.

Cate fructe maximum incap intr-un cos? Cate cosuri de fiecare?

Rezolvare:

- -

- -

- -

- -

Max fructe/cos = ___ ; cosuri mere = ___ ; cosuri pere = ___

b) Doua autobuze pleaca simultan. Unul la fiecare 18 minute,

celalalt la fiecare 24 de minute. Dupa cat timp pleaca din nou impreuna?

Rezolvare:

- -

- -

- -

Raspuns: dupa ___ minute

c) Gaseste cel mai mic numar de 4 cifre divizibil cu 6, 8 si 9.

Rezolvare:

- -

- -

- -

- -

Raspuns: ___

d) O cutie contine carti rosii, albastre si galbene in raport 2:3:5.

Total = 120 carti. Cate sunt de fiecare culoare?

- -

- -

Rosii = ___ ; Albastre = ___ ; Galbene = ___

BONUS (10 puncte)

Ex. B1. Demonstreaza ca produsul oricaror 3 numere naturale consecutive este divizibil cu 6. (4p)

(Indicatie: $n(n+1)(n+2)$ - arata ca e div. cu 2 si cu 3.)

Ex. B2. Fie $n = 2^a \times 3^b$ cu $a, b > 0$. (3p)

Stiind ca n are 12 divizori si $n < 200$, determina n.

(nr. div = $(a+1)(b+1) = 12$; incearca combinatiile posibile)

n = ___

Ex. B3. Gaseste toate perechile (a, b) cu $a \leq b$, (3p)

c.m.m.d.c.$(a,b) = 6$ si c.m.m.m.c.$(a,b) = 180$.

Perechile: ____________________

Barem de corectare

Ex. 1	12p	a-h) 1.5p fiecare (8 x 1.5p)
Ex. 2	8p	a) 4p b) 4p (0.5p fiecare)
Ex. 3	10p	a) 4x2p=8p b) 2p
Ex. 4	15p	a) 3p b) 4x2p=8p c) 4p
Ex. 5	15p	a) 5p b) 5p c) 5p
Ex. 6	15p	a) 3p b) 5p c) 3p d) 4p
Ex. 7	10p	a) 5x1.5p b) 2.5p
Ex. 8	15p	a) 4p b) 4p c) 4p d) 3p